

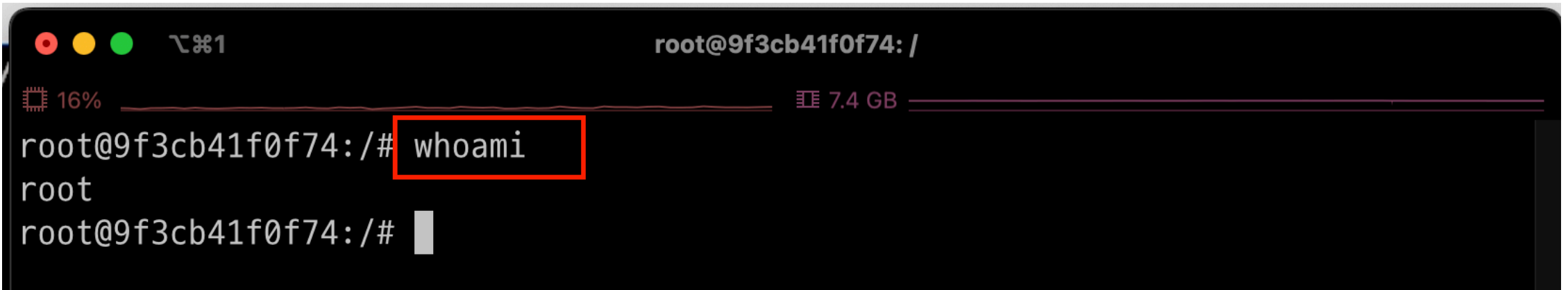
기본명령어

- 리눅스 명령어는 결국 셸이 제공하는 명령어임
- 따라서 리눅스 기본 명령어는 bash에서 제공하는 명령어임

단계1: whoami

- 로그인한 사용자ID를 알려줌

```
whoami
```

A terminal window with a dark background. The title bar shows three colored circles (red, yellow, green) and the text 'ㄸ₩1'. The terminal prompt is 'root@9f3cb41f0f74: /'. Below the prompt, the command 'whoami' is entered and highlighted with a red rectangular box. The output of the command is 'root', displayed on the line immediately following the command. The terminal also shows system status indicators at the top: a battery icon with '16%' and a memory icon with '7.4 GB'.

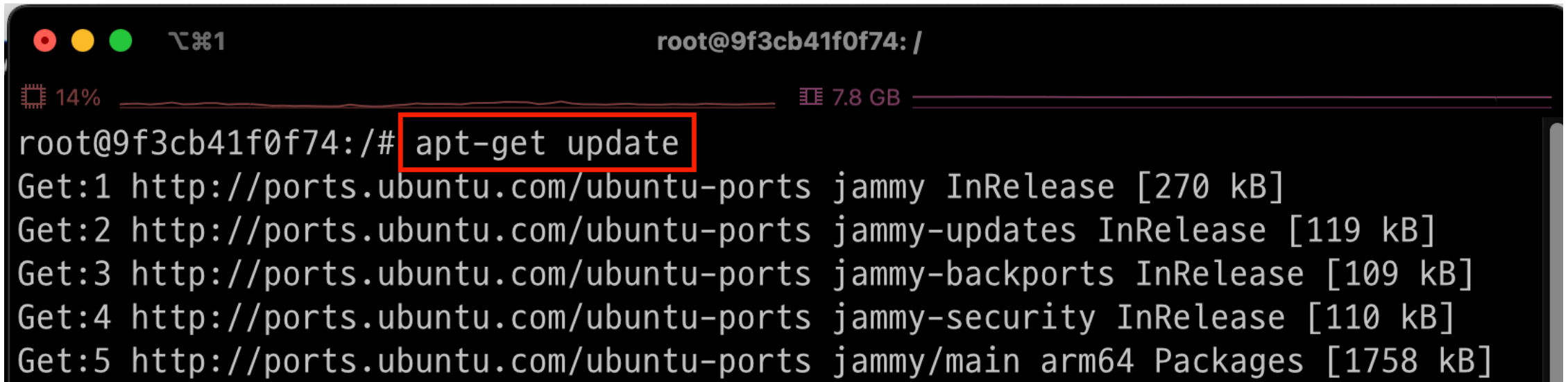
```
root@9f3cb41f0f74: /  
root@9f3cb41f0f74: /# whoami  
root  
root@9f3cb41f0f74: /#
```

단계2: apt / apt-get

- 우분투(Ubuntu)를 포함한 데비안(Debian)계열의 리눅스에서 쓰이는 패키지 관리 명령어 도구입니다.
- apt를 쓰든 apt-get(또는 apt-cache)을 쓰든 내부 동작의 차이는 거의 없습니다.

- 업데이트할 패키지들의 목록을 최신으로 갱신한다. 실제로 패키지를 업그레이드 하는게 아니다.

```
apt-get update
```

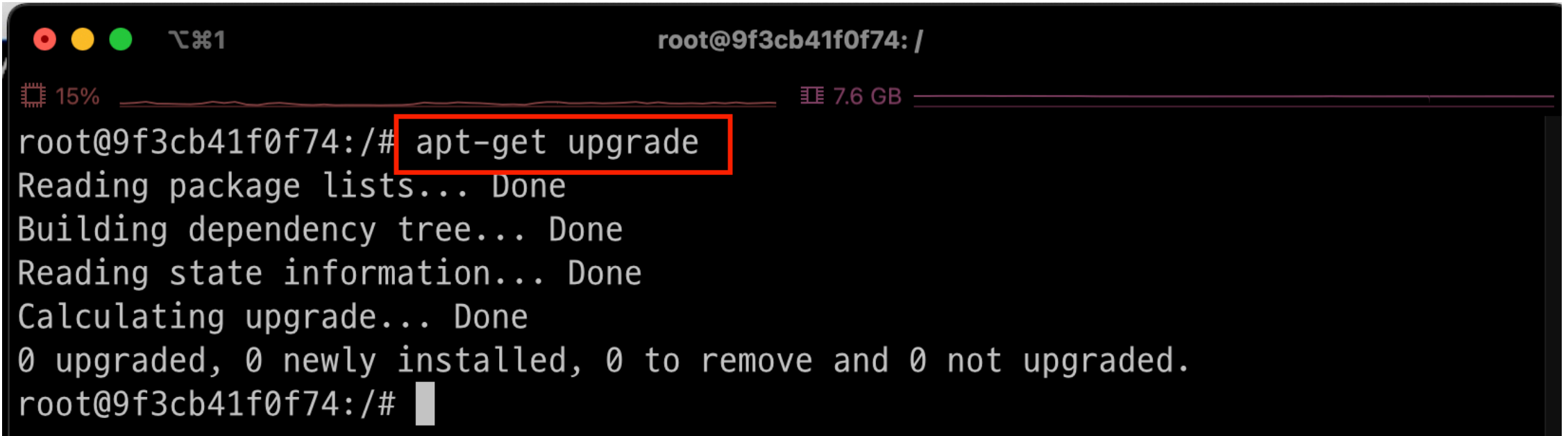


A terminal window screenshot with a dark background. The title bar shows three colored circles (red, yellow, green) and the text 'root@9f3cb41f0f74: /'. The terminal content shows the command 'apt-get update' being executed, with the command text highlighted by a red rectangular box. Below the command, the output shows five lines of package information being retrieved from 'http://ports.ubuntu.com/ubuntu-ports' for the 'jammy' release, including 'InRelease' files and 'Packages' for the 'arm64' architecture.

```
root@9f3cb41f0f74: /# apt-get update
Get:1 http://ports.ubuntu.com/ubuntu-ports jammy InRelease [270 kB]
Get:2 http://ports.ubuntu.com/ubuntu-ports jammy-updates InRelease [119 kB]
Get:3 http://ports.ubuntu.com/ubuntu-ports jammy-backports InRelease [109 kB]
Get:4 http://ports.ubuntu.com/ubuntu-ports jammy-security InRelease [110 kB]
Get:5 http://ports.ubuntu.com/ubuntu-ports jammy/main arm64 Packages [1758 kB]
```

- update 로 갱신된 최신 패키지들을 실제로 업그레이드 한다.

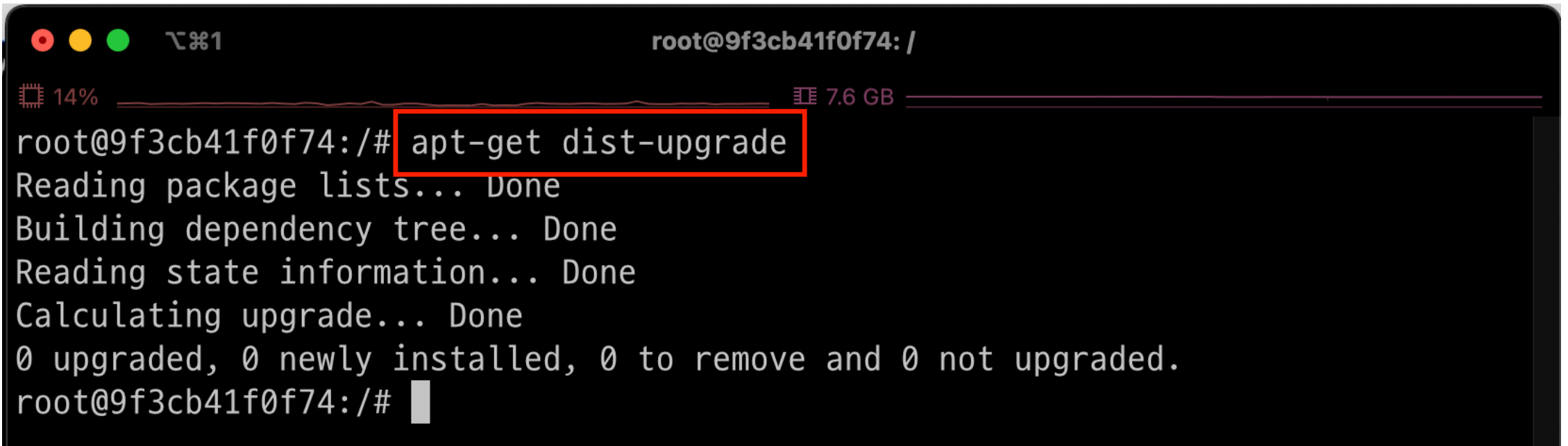
```
apt-get upgrade
```

A terminal window with a dark background. The title bar shows three colored circles (red, yellow, green) and the text 'ㄸ₩1'. The terminal prompt is 'root@9f3cb41f0f74: /'. The command 'apt-get upgrade' is entered and highlighted with a red box. The output shows the package lists being read, the dependency tree being built, and state information being read, all successfully. It then calculates the upgrade and reports that 0 packages are upgraded, 0 are newly installed, 0 are to be removed, and 0 are not upgraded. The prompt returns to 'root@9f3cb41f0f74: /#'.

```
root@9f3cb41f0f74: /# apt-get upgrade
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Calculating upgrade... Done
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
root@9f3cb41f0f74: /#
```

- upgrade 는 의존성 체크를 하지 않고 업그레이드하지만 dist-upgrade는 의존성 체크를 하면서 위의 upgrade에서 수행되지 못한 dependency 까지 설치해준다.

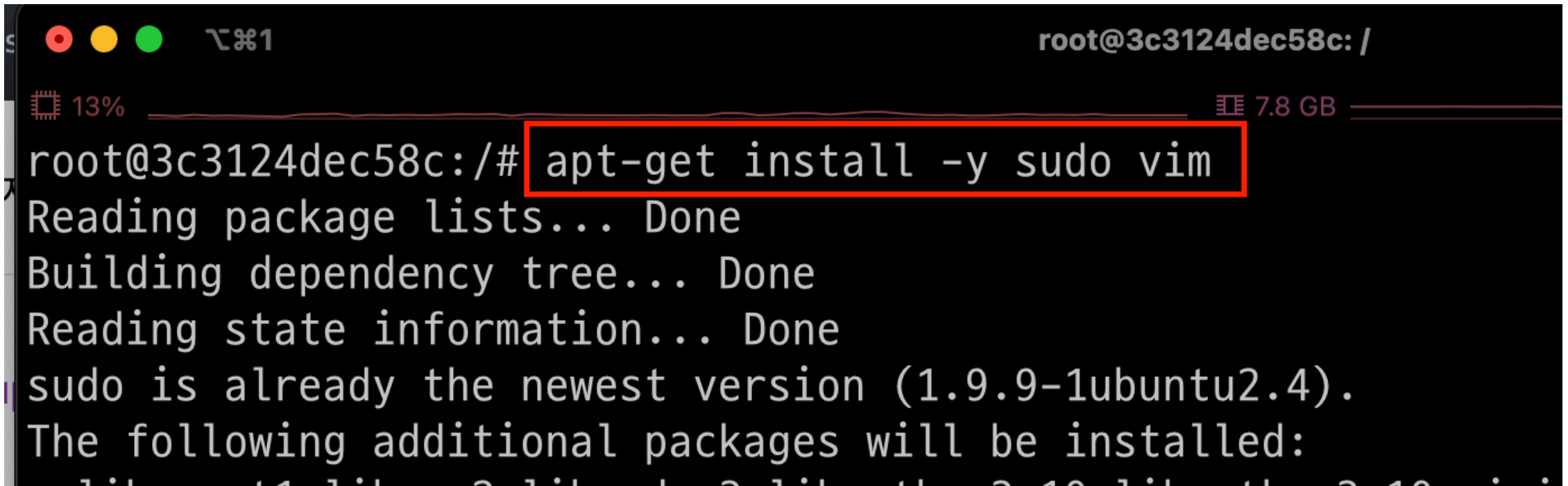
```
apt-get dist-upgrade
```

A terminal window with a dark background. The title bar shows three colored circles (red, yellow, green) and a keyboard icon. The terminal prompt is root@9f3cb41f0f74: /. The command apt-get dist-upgrade is entered and highlighted with a red box. The output shows the package lists being read, the dependency tree being built, and state information being read, all successfully. It then calculates the upgrade, showing 0 packages to be upgraded, installed, removed, or not upgraded. The prompt returns to root@9f3cb41f0f74: /#.

```
root@9f3cb41f0f74: /# apt-get dist-upgrade
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Calculating upgrade... Done
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
root@9f3cb41f0f74: /#
```

- 패키지/라이브러리 설치
- 명령어: apt-get install -y <라이브러리명>
 - `sudo` : 일반사용자에게 root 권한을 부여
 - `vim` : 리눅스용 에디터

```
apt-get install -y sudo vim
```

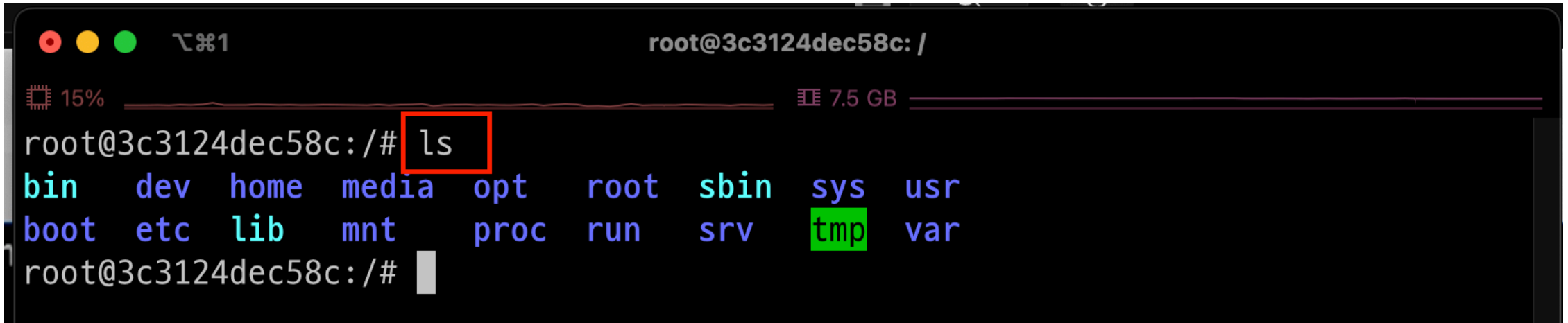


```
root@3c3124dec58c: /  
13% 7.8 GB  
root@3c3124dec58c: /# apt-get install -y sudo vim  
Reading package lists... Done  
Building dependency tree... Done  
Reading state information... Done  
sudo is already the newest version (1.9.9-1ubuntu2.4).  
The following additional packages will be installed:
```

단계3: ls

- 파일 목록 출력

```
ls
```

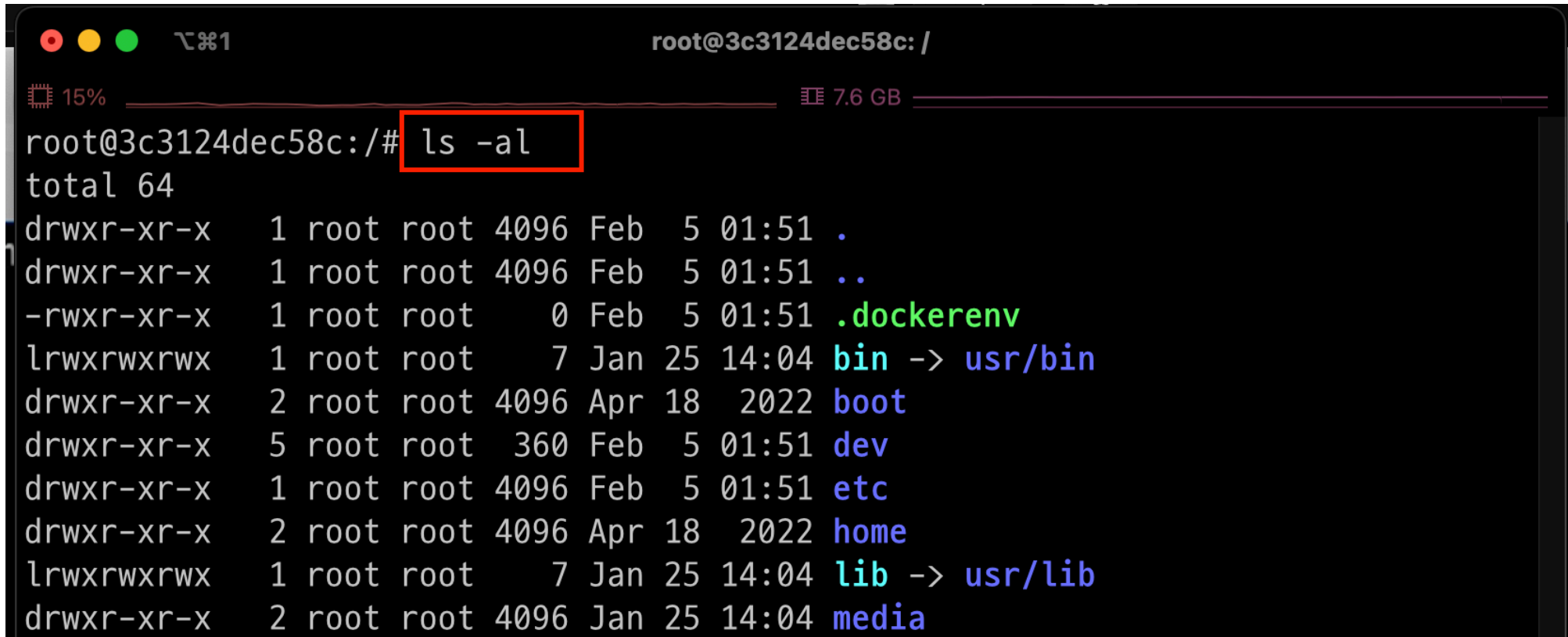


A terminal window with a dark background. The title bar shows three colored circles (red, yellow, green) and the text 'ㄸ%1'. The terminal content shows the prompt 'root@3c3124dec58c: /' followed by the command 'ls' which is highlighted with a red box. The output of the command is displayed in two rows of colored text: 'bin dev home media opt root sbin sys usr' on the first row and 'boot etc lib mnt proc run srv tmp var' on the second row. The 'tmp' directory is highlighted with a green background. Below the output, the prompt 'root@3c3124dec58c: /#' is shown with a cursor.

```
root@3c3124dec58c: /  
15% 7.5 GB  
root@3c3124dec58c: /# ls  
bin dev home media opt root sbin sys usr  
boot etc lib mnt proc run srv tmp var  
root@3c3124dec58c: /#
```

- `-a` : 모든 파일 표시
- `-l` : 파일에 관한 상세 정보 표시

```
ls -al
```

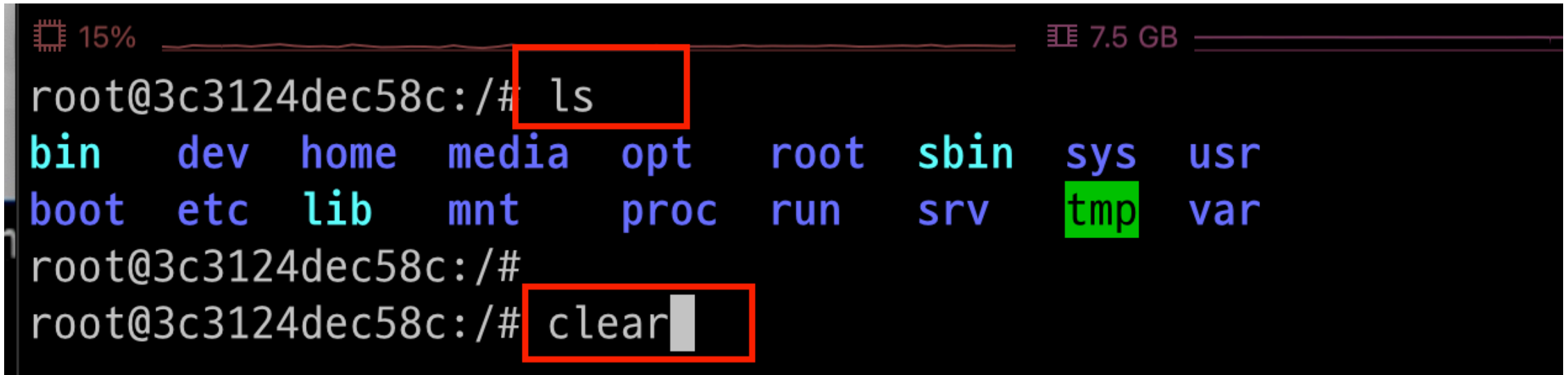


```
root@3c3124dec58c: /  
15% 7.6 GB  
root@3c3124dec58c: /# ls -al  
total 64  
drwxr-xr-x  1 root root 4096 Feb  5 01:51 .  
drwxr-xr-x  1 root root 4096 Feb  5 01:51 ..  
-rwxr-xr-x  1 root root    0 Feb  5 01:51 .dockerenv  
lrwxrwxrwx  1 root root    7 Jan 25 14:04 bin -> usr/bin  
drwxr-xr-x  2 root root 4096 Apr 18 2022 boot  
drwxr-xr-x  5 root root  360 Feb  5 01:51 dev  
drwxr-xr-x  1 root root 4096 Feb  5 01:51 etc  
drwxr-xr-x  2 root root 4096 Apr 18 2022 home  
lrwxrwxrwx  1 root root    7 Jan 25 14:04 lib -> usr/lib  
drwxr-xr-x  2 root root 4096 Jan 25 14:04 media
```

단계4: clear

- 터미널 화면을 깨끗하게

```
clear
```



A terminal window screenshot with a dark background. At the top, there are status indicators: a battery icon with '15%' and a memory icon with '7.5 GB'. The prompt is 'root@3c3124dec58c: /#'. The first command entered is 'ls', which is highlighted with a red box. The output shows a list of directories: 'bin', 'dev', 'home', 'media', 'opt', 'root', 'sbin', 'sys', 'usr' on the first line, and 'boot', 'etc', 'lib', 'mnt', 'proc', 'run', 'srv', 'tmp', 'var' on the second line. The 'tmp' directory is highlighted with a green box. The second prompt is 'root@3c3124dec58c: /#'. The third command entered is 'clear', which is highlighted with a red box.

```
root@3c3124dec58c: /# ls
bin  dev  home  media  opt  root  sbin  sys  usr
boot  etc  lib  mnt  proc  run  srv  tmp  var
root@3c3124dec58c: /#
root@3c3124dec58c: /# clear
```

단계5: cd

- 디렉토리 이동
- 명령어: cd <디렉토리 path>
 - cd ~ : 해당 계정의 홈 디렉토리로 이동
 - cd .. : 상위 디렉토리로 이동

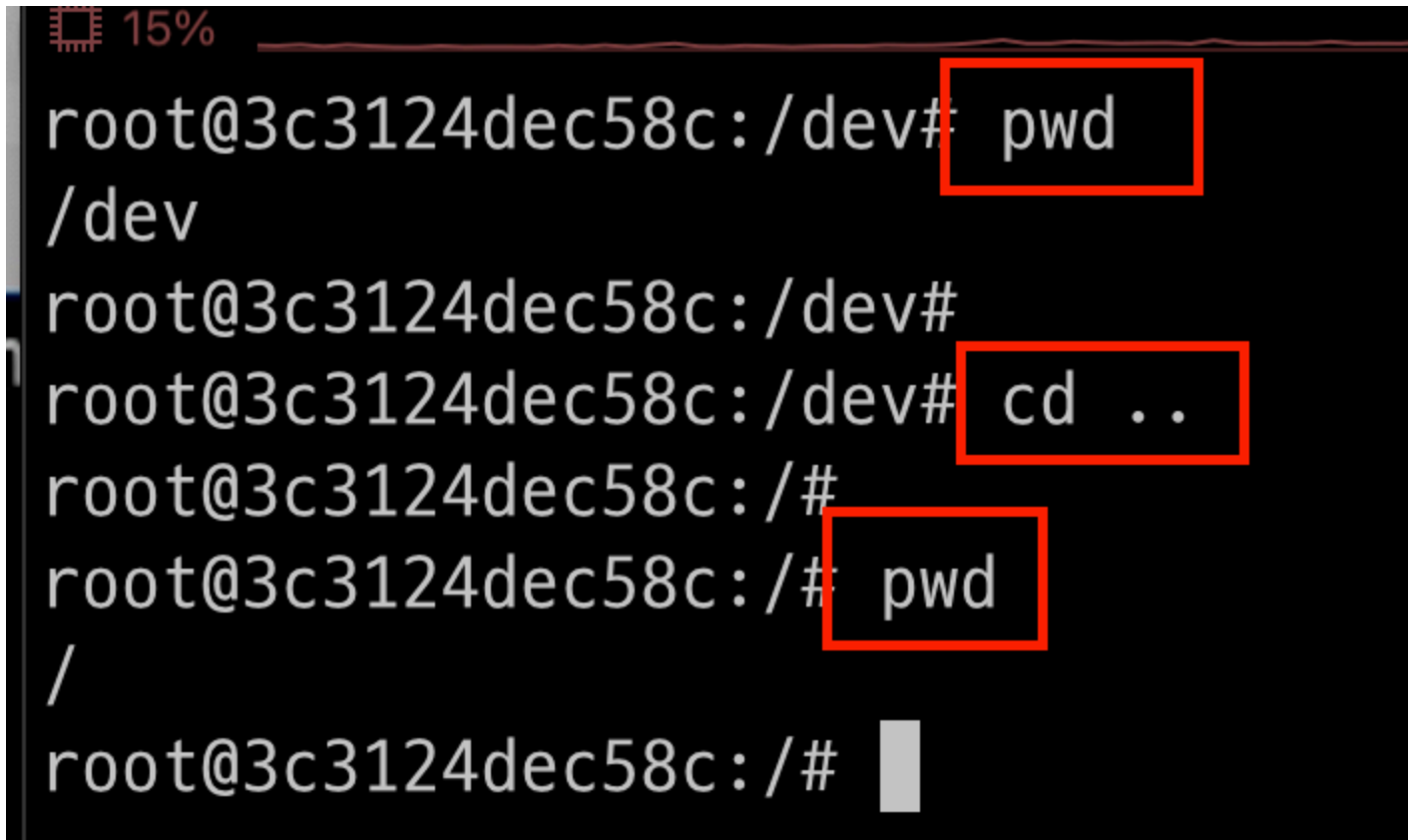
cd dev

```
root@3c3124dec58c:/# ls
bin  dev  home  media  opt  root  sbin  sys  usr
boot etc  lib   mnt    proc run   srv   tmp  var
root@3c3124dec58c:/#
root@3c3124dec58c:/# cd dev
root@3c3124dec58c:/dev#
root@3c3124dec58c:/dev# ls
console  fd      mqueue  ptmx  random  stderr  stdout  urandom
core     full    null    pts   shm      stdin   tty     zero
root@3c3124dec58c:/dev#
```

단계6: pwd

- 현재 디렉토리 위치

pwd

A terminal window with a black background and white text. At the top left, there is a battery icon and '15%'. The terminal shows a sequence of commands and their outputs. The first command is 'pwd' in the prompt 'root@3c3124dec58c: /dev#', which is highlighted with a red box. The output is '/dev'. The second command is 'cd ..' in the prompt 'root@3c3124dec58c: /dev#', also highlighted with a red box. The output is 'root@3c3124dec58c: /#'. The third command is 'pwd' in the prompt 'root@3c3124dec58c: /#', highlighted with a red box. The output is '/'. The final line shows the prompt 'root@3c3124dec58c: /#' with a cursor.

```
15%  
root@3c3124dec58c: /dev# pwd  
/dev  
root@3c3124dec58c: /dev#  
root@3c3124dec58c: /dev# cd ..  
root@3c3124dec58c: /#  
root@3c3124dec58c: /# pwd  
/  
root@3c3124dec58c: /#
```

단계7: cat

- 파일내용출력
- 명령어: cat <파일명>

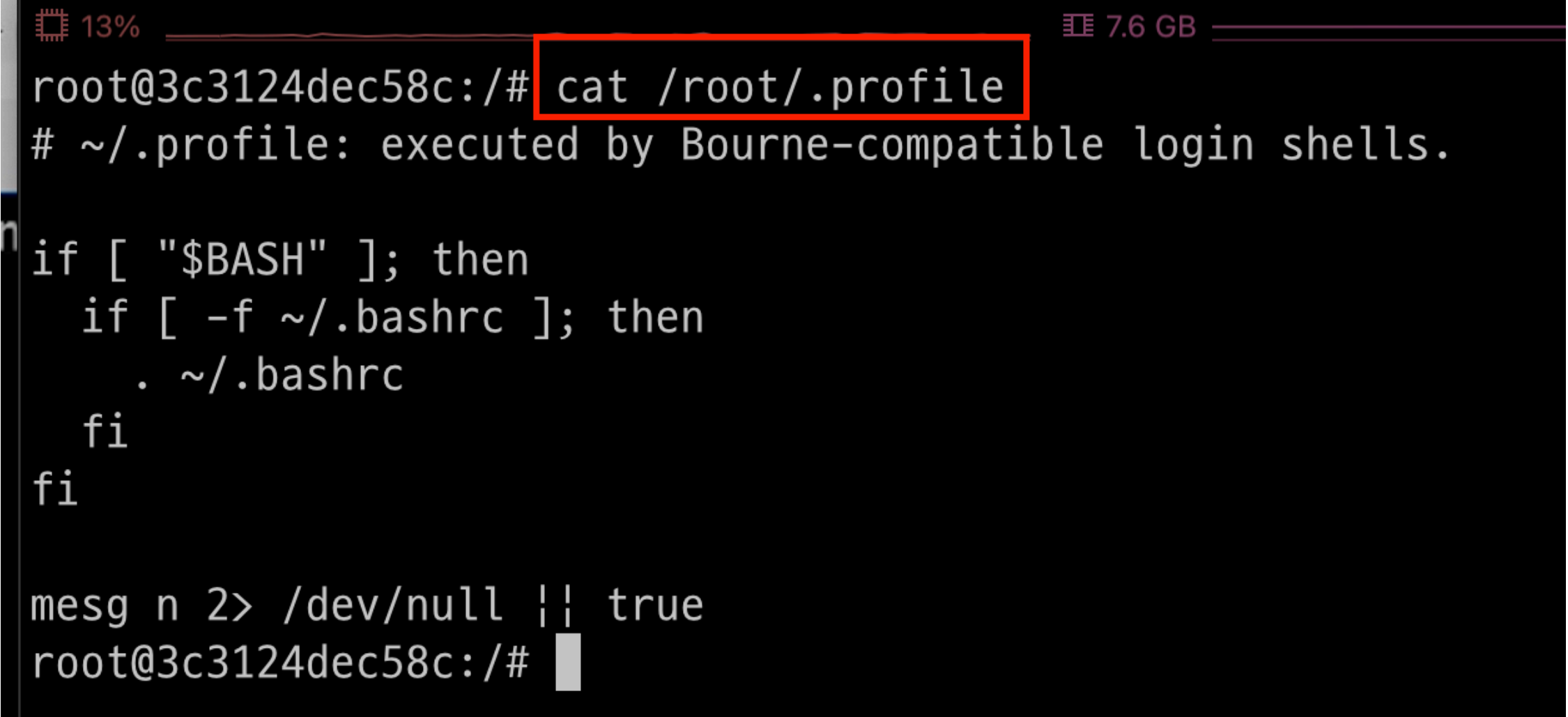
```
cat .profile
```

```
13% 7.6 GB
root@3c3124dec58c:/# cd ~
root@3c3124dec58c:~#
root@3c3124dec58c:~# ls -al
total 16
drwx----- 2 root root 4096 Jan 25 14:09 .
drwxr-xr-x 1 root root 4096 Feb  5 01:51 ..
-rw-r--r-- 1 root root 3106 Oct 15  2021 .bashrc
-rw-r--r-- 1 root root  161 Jul  9  2019 .profile
root@3c3124dec58c:~#
root@3c3124dec58c:~# cat .profile
# ~/.profile: executed by Bourne-compatible login shells.

if [ "$BASH" ]; then
  if [ -f ~/.bashrc ]; then
    . ~/.bashrc
  fi
fi

mesg n 2> /dev/null || true
root@3c3124dec58c:~#
```

```
cat /root/.profile
```



A terminal window with a dark background. The top status bar shows a CPU icon, '13%', a memory icon, and '7.6 GB'. The prompt is 'root@3c3124dec58c:/#'. The command 'cat /root/.profile' is entered and highlighted with a red box. The output shows the contents of the profile file, including a comment about Bourne-compatible login shells and a conditional block for Bash that sources ~/.bashrc. The terminal ends with 'mesg n 2> /dev/null || true' and a new prompt.

```
root@3c3124dec58c:/# cat /root/.profile
# ~/.profile: executed by Bourne-compatible login shells.

if [ "$BASH" ]; then
  if [ -f ~/.bashrc ]; then
    . ~/.bashrc
  fi
fi

mesg n 2> /dev/null || true
root@3c3124dec58c:/#
```

단계8: mkdir

- 폴더 생성

```
mkdir sample
```

```
14%  
root@3c3124dec58c:~# cd ~  
root@3c3124dec58c:~#  
root@3c3124dec58c:~# ls  
root@3c3124dec58c:~#  
root@3c3124dec58c:~# mkdir sample  
root@3c3124dec58c:~#  
root@3c3124dec58c:~# ls  
sample  
root@3c3124dec58c:~#
```

- `-p` : 부모폴더가 없다면 같이 생성

```
mkdir -p /root/tutorial/test
```

```
root@3c3124dec58c:~# pwd
/root
root@3c3124dec58c:~# ls
sample
root@3c3124dec58c:~# mkdir /root/tutorial/test
mkdir: cannot create directory '/root/tutorial/test': No such file or directory
root@3c3124dec58c:~#
root@3c3124dec58c:~# mkdir -p /root/tutorial/test
root@3c3124dec58c:~#
root@3c3124dec58c:~# ls
sample tutorial
root@3c3124dec58c:~#
```

단계9: touch

- 빈 파일 생성
- 명령어: touch <파일명>

```
touch test.txt
```

```
root@3c3124dec58c:~# pwd
/root
root@3c3124dec58c:~# ls
sample tutorial
root@3c3124dec58c:~# touch test.txt
root@3c3124dec58c:~#
root@3c3124dec58c:~# ls
sample test.txt tutorial
root@3c3124dec58c:~#
```

단계10: mv

- 이름변경 또는 이동
- 명령어: mv <이동할 파일> <이동될 파일>

이름변경

```
mv test.txt test1.txt
```

```
root@3c3124dec58c:~# ls
sample test.txt tutorial
root@3c3124dec58c:~#
root@3c3124dec58c:~# mv test.txt test1.txt
root@3c3124dec58c:~#
root@3c3124dec58c:~# ls
sample test1.txt tutorial
root@3c3124dec58c:~#
```

이동

```
mv test1.txt /root/sample/test1.txt
```

ls # 기존 파일 확인 > 없음

14%

7.8 GB

```
root@3c3124dec58c:~# mv test1.txt /root/sample/test1.txt
```

```
root@3c3124dec58c:~#
```

```
root@3c3124dec58c:~# ls
```

```
sample tutorial
```

```
root@3c3124dec58c:~# cd sample/
```

```
root@3c3124dec58c:~/sample#
```

```
root@3c3124dec58c:~/sample# ls
```

```
test1.txt
```

```
root@3c3124dec58c:~/sample#
```

단계11: cp

- 파일이나 디렉토리를 복사
- 명령어: cp <복사할 파일> <복사될 파일>

```
cp test1.txt /root/test1.txt
ls # 기존 파일 확인
ls /root # 복사한 파일 확인
```

```
root@3c3124dec58c:~/sample# ls
test1.txt
root@3c3124dec58c:~/sample# cp test1.txt /root/test1.txt
root@3c3124dec58c:~/sample# ls
test1.txt
root@3c3124dec58c:~/sample# ls /root
sample test1.txt tutorial
root@3c3124dec58c:~/sample#
```

- `-r` : 하위 디렉토리 까지 모두 복사합니다.

```
cp -r /root/sample /root/tutorial
ls /root/tutorial/sample
```

```
root@3c3124dec58c:~/tutorial# ls
test
root@3c3124dec58c:~/tutorial# cp -r /root/sample /root/tutorial
root@3c3124dec58c:~/tutorial# ls
sample test
root@3c3124dec58c:~/tutorial# ls /root/tutorial/sample
test1.txt
root@3c3124dec58c:~/tutorial#
```

단계12: rm

- 파일이나 디렉토리 삭제
- 명령어: rm <삭제할 파일>

```
rm test1.txt
```

```
root@3c3124dec58c:~# ls
sample test1.txt tutorial
root@3c3124dec58c:~#
root@3c3124dec58c:~# rm test1.txt
root@3c3124dec58c:~#
root@3c3124dec58c:~# ls
sample tutorial
root@3c3124dec58c:~#
```

- `-r`: 디렉토리 삭제

```
rm -r sample
```

```
root@3c3124dec58c:~# ls
sample tutorial
root@3c3124dec58c:~# rm sample
rm: cannot remove 'sample': Is a directory
root@3c3124dec58c:~#
root@3c3124dec58c:~# rm -r sample
root@3c3124dec58c:~#
root@3c3124dec58c:~# ls
tutorial
root@3c3124dec58c:~#
```